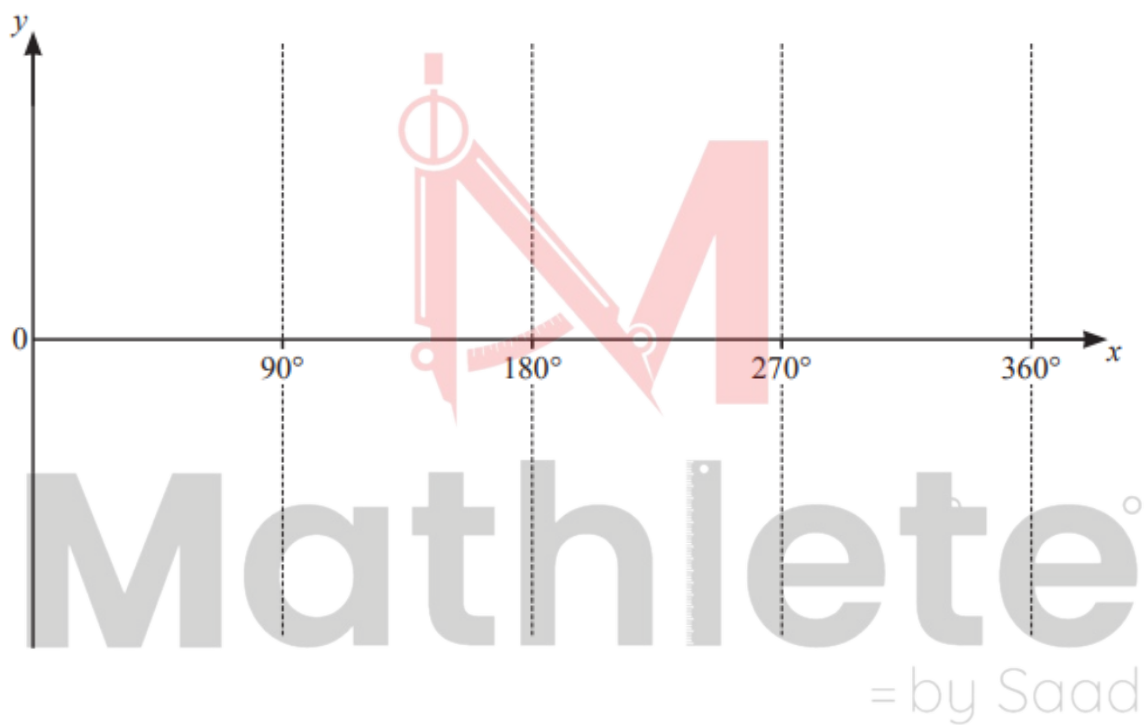


Trigonometry

Worksheet 5

Trigonometric Graphs and Equations

1. W20/qp41/q10(b)



- i. On the diagram, sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$. [2]
- ii. Solve the equation $5 \tan x = -7$ for $0^\circ \leq x \leq 360^\circ$. [3]

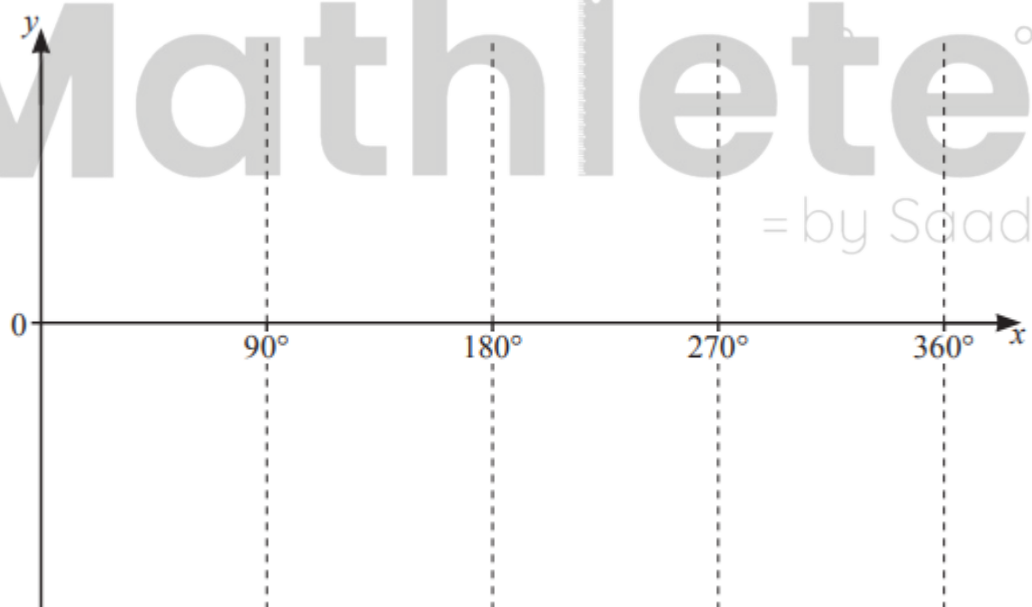
2. W22/qp42/q9



- a. On the diagram, sketch the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$. [2]
- b. Solve the equation $5 \sin x + 4 = 0$ for $0^\circ \leq x \leq 360^\circ$. [3]

3. M23/qp42/q12

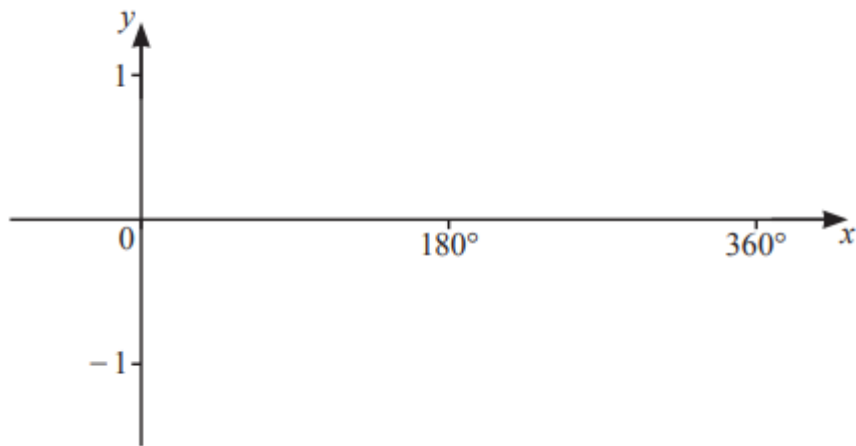
- a. Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$. [2]



- b. Find x when $\tan x = \frac{1}{\sqrt{3}}$ and $0^\circ \leq x \leq 360^\circ$. [2]

4. S23/qp43/q7(c)

i.



On the diagram, sketch the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$. [2]

ii. Solve the equation $3 \sin x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$. [3]

Answer Key

1. W20/qp41/q10(b)

10(b)(i)	Correct sketch	2 B1 for one correct section out of 4 OR B1 for two properties correct from • Crosses x-axis at $(0, 0)$, $(180, 0)$ and $(360, 0)$ only • Correct curvature in each section of 90° • Asymptotes at $x = 90$ and $x = 270$
10(b)(ii)	125.5 or 125.53 to 125.54 and 305.5 or 305.53 to 305.54	3 B2 for one correct angle or B1 for -54.5 or $-54.46\dots$ or for 2 angles with a difference of 180.

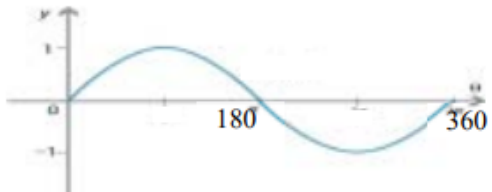
2. W22/qp42/q9

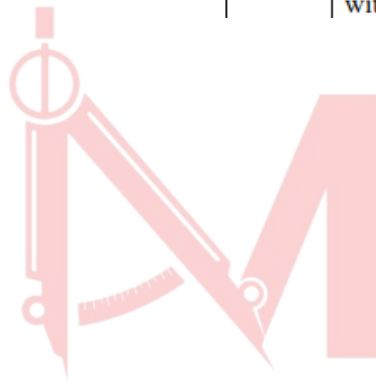
9(a)	Correct sketch to go through $(0, 0)$, and $(360, 0)$	2 M1 for correct sine curve shape through the origin or for almost correct sketch fitting all tramlines but with an omission at either end or incorrect curvature in one place only
9(b)	233.1 or 233.13... and 306.9 or 306.86 to 306.87	3 B2 for one correct angle or M1 for $\sin x = -0.8$ oe If 0 scored SC1 for 2 reflex angles that add to 540 or two non-reflex angles that add to 180

3. M23/qp42/q12

12(a)	Correct sketch	2 Condone curve touching asymptotes but not crossing B1 for one section correct or for 3 sections in correct part of graph but with incorrect curvature and no other sections in incorrect part of graph
12(b)	30 and 210 final answer	2 B1 for each If 0 scored SC1 for two answers (one acute and one reflex) with a difference of 180

4. S23/qp43/q7(c)

7(c)(i)	Correct sketch through (0, 0) (180, 0) and (360, 0) with max and min at 1 and -1 resp. 	2 B1 for correct sine curve shape, starting at the origin, with minimum of 1 cycle.
7(c)(ii)	199.5 or 199.47... and 340.5...	3 B2 for one correct or M1 for $\sin x = -\frac{1}{3}$ oe If 0 scored, SC1 for two reflex angles with a sum of 540 or 2 non-reflex angles with a sum of 180



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